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Miniproteins are the TikZ unicorn of biology

4.5k words, 19-24 min reading time



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SEP 03, 2026



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Miniprotein binders have become the easiest impressive thing that you can do in protein AI today. If you don't believe me, look at the news. In August 2026, a team at Anthropic demonstrated that Claude could [autonomously design miniproteins for 14 out of 15 human targets](#). The work was led by [Nathan Frey](#), a very well-respected AI for Bio researcher, and formerly one of the most visible leaders at [Prescient Design](#). The announcement has received a lot of (imho, valid) criticism. Some argue that the designs are not made directly by Claude, but instead by dozens of open source software tools that [aren't mentioned even once in the blog](#), and that the [\(massive!\) prompt](#) doing the orchestrating is [clearly written by humans with incredibly sophisticated expertise](#). Others have attacked Anthropic's claim that "*designing a binder is an easier process than designing a drug, but it's a useful proxy*", arguing that it is at best the first step of a much longer and harder process.

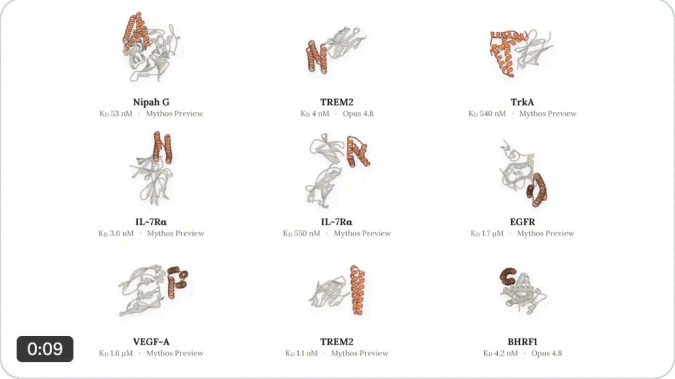
I agree with both stances. But there's a subtle, critical point that most commenters have left behind: **can miniprotein binders ever be transformed into drugs?**

AI Anthropic @AnthropicAI · Aug 18

Many drugs work by binding to a specific target in the body and blocking or changing what it does. An important first step in the drug development process is designing a molecule that can bind tightly to its target. Traditionally, that's meant weeks or months of expert work per target, sifting through a large number of candidates to identify the few that work.

We wanted to test if Claude could successfully design novel protein binders from scratch (also called de novo design). With a protein design prompt written by a human expert, Claude autonomously designed protein binders against 14 out of 15 targets.

We then worked with Adaptyv Bio and Twist Bioscience, who independently built and tested the proteins Claude designed.



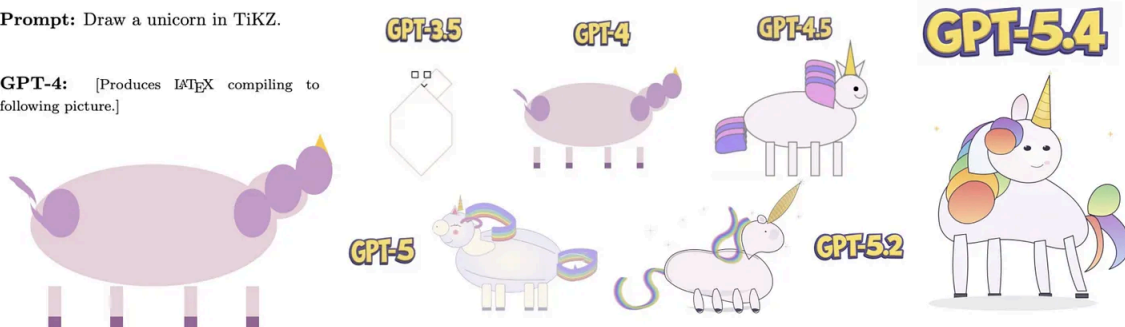
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The team described that “designing a binder is an easier process than designing a drug, but it’s a useful proxy”.

A few years ago, none of this was possible. The emergence of open source deep learning models that can reach 1-10% success rates, coupled with cloud laboratories like [Adaptyv](#), has enabled a new generation of protein designers who never need to lift a pipette. Democratising life sciences research is overwhelmingly a good thing¹, but frankly, look at what this new capability is mostly being used for. In recent years, nearly every protein AI model has been released with a list of miniprotein binders designed and validated at a cloud lab. We have come to make miniprotein binder design a benchmark that is flashy and attractive, but also a proxy metric that can be optimised on, and that is not necessarily reflective of the model’s true capabilities. Beware: miniproteins have become the *TikZ unicorn of biology*.

Prompt: Draw a unicorn in TikZ.

GPT-4: [Produces L^AT_EX compiling to following picture.]



The TikZ unicorn benchmark was proposed in the celebrated [Sparks of AGI](#) paper, by Microsoft Research, shown on the left. The idea of the benchmark is to ask LLMs to generate the image of a unicorn in the complicated vector graphics language

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TikZ, as a way to simultaneously test semantic understanding, spatial reasoning and programming syntax. The image on the right is reproduced from a tweet by Sébastien Bubeck, the first author of the paper.

The silent claim is that these miniprotein binders could be The Next Big Thing™ in drug discovery. At a cursory glance, you might be convinced. [ProteinBase](#), the free database generated by Adaptiv showcases hundreds of examples against valuable human targets: [MDM2](#), the protein that degrades [p53](#) (the *guardian of the genome*); [EGFR](#), the growth factor receptor behind proliferation in many cancers; [IL-7Rα](#), one of the darlings of the *immunology and inflammation* (I&I) boom. These are all clinically relevant targets, but as soon as you probe closely they start to fall apart. Many of these proteins are intracellular and cannot be targeted with a miniprotein binder. The rest are either well-served (EGFR inhibitors abound) or have clear on-target toxicity (MDM2 inhibitors have been unsuccessful in the clinic). What these targets do share is that they are all well-characterised, structurally tractable, and eminently assayable. The questions that tell you whether these miniproteins could ever be given to a patient (translatability, pharmacokinetics) are never addressed. This is the nature of the benchmark: nobody actually needs a TikZ unicorn.

Whether miniprotein binders can ever become good drugs is not a trivial question. The default answer in the field is that, since clinical development typically takes 10-15 years from discovery to approval, we simply don't have the information. What gets lost is that designed miniproteins have been tried and tested for decades. At least a dozen biotech companies have made designed proteins the centre of their strategy. While most of them are not AI-designed (phage display, rational engineering, etc), the difficulties that these molecules ran into have nothing to do with their sequence. Instead, they stem from properties of proteins themselves like their size, inability to cross a membrane, or tendency to be filtered out by the kidney. Which means, of course, that we have a pretty good idea of how they will play out in the clinic.

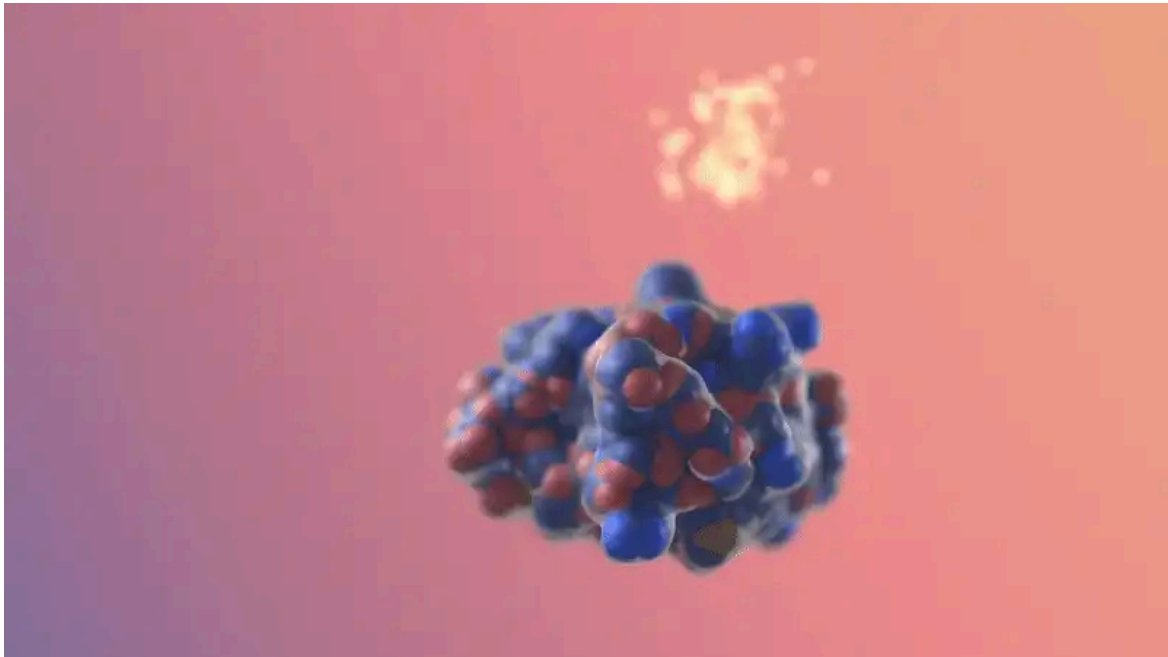
In this essay, I would like to make a point that designed miniprotein binders are not the “almost ready” drugs that some are trying to paint, but that have many liabilities that make them generally bad drugs. To make the case, I will:

- look at some of the miniprotein binders (affibodies, DARPins, etc) that have spent the last decade in the clinic, and ask why their success has been limited
- examine how the companies built on them are positioning themselves today, and where the whole field seems to be going
- make a few predictions about the future of miniprotein binders, and how drug discovery is likely not where they succeed

What are miniprotein binders

Miniprotein binders (from now on, *just* miniproteins) are the cheap, stable, small alternative to our beloved antibodies... or, at least, that is the pitch. More accurately, miniproteins are small proteins (duh) that bind other proteins with strong affinity. They are typically under 200 amino acids, and most commonly around 60 to 100 amino acids², whereas an antibody runs to about 1,300.

Miniproteins are, practically speaking, an antibody's binding site without the antibody parts baked in. You get the specificity, and you get very high binding affinity, typically nanomolar to picomolar. Unfortunately, you also get the constraint that comes with it. A protein is a large, polar, charged object, and a cell membrane is a sheet of lipid: there is no passive route across it at any size, which is why antibodies cannot get into cells either. Miniproteins cannot permeate the cell, ruling out all the intracellular protein targets. This is not only most of the proteome, but also the fraction that is hardest to drug! It also means that most miniproteins are blockers rather than agonists.

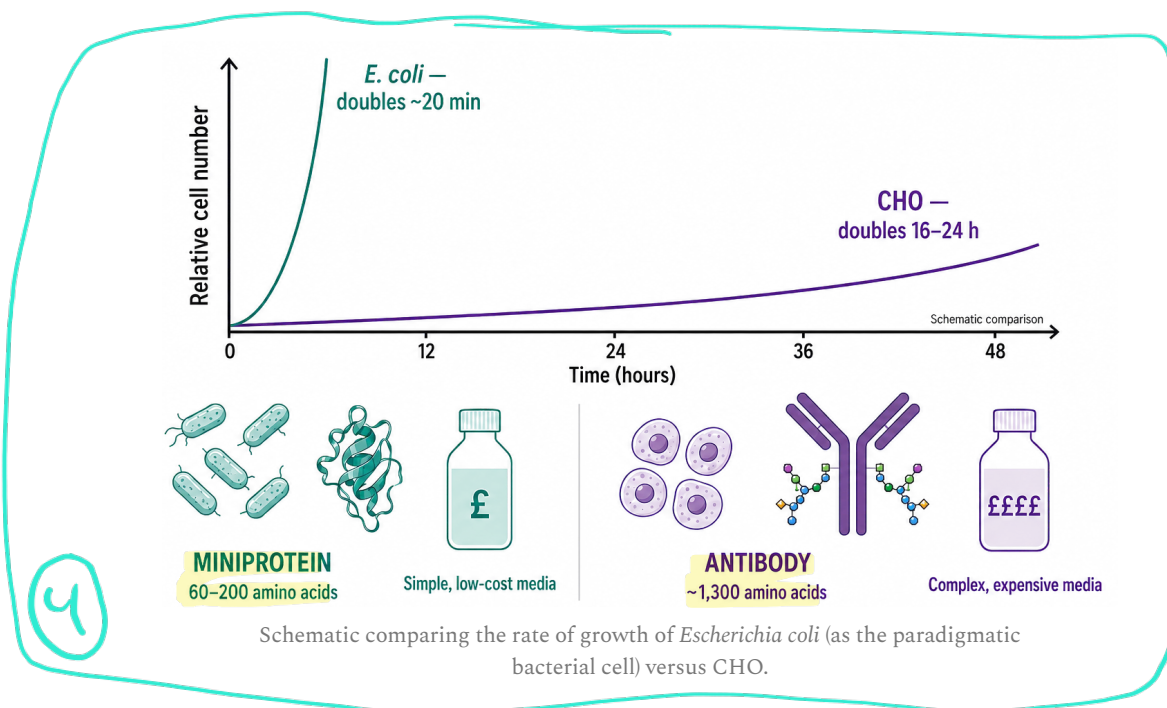


3 Animation ([source](#)) showing a miniprotein being designed by RFdiffusion to bind the insulin receptor.

So, why bother with miniproteins, when antibodies already exist? First of all, stability. Antibodies are a struggle to keep stable, and they generally require refrigeration. Miniproteins routinely achieve very high stabilities, and there are now many reports of [miniproteins that survive for years on a shelf](#). They can even be made to be [pH- and protease-resistant](#), which opens the possibility of oral administration, unthinkable in an antibody.

The bigger advantage is in production: miniproteins can generally be made in bacterial cells. Antibody therapies, with the exception of a few fragments and nanobodies, need

to be expressed in a mammalian cell, because of restrictions in folding and post-translational modifications. A [Chinese Hamster Ovary \(CHO\) cell](#), typically used as a mammalian expression system, doubles every 16-24 hours; and nearly every ingredient, from media to the bioreactor's culture bag, is extremely expensive. Miniproteins can be expressed in *Escherichia coli*, which doubles about once every 20 minutes, and grows in very economical media. It is no surprise that alternatives such as affibodies or DARPin have arisen, because they offer many of the advantages of the antibodies but can be produced much more easily. And it's not just cost³. One of the big problems of developing a drug is making sure that it can be produced at scale under [Good Manufacturing Practices](#). When the production becomes faster and cheaper, your timeline to the clinic shortens.



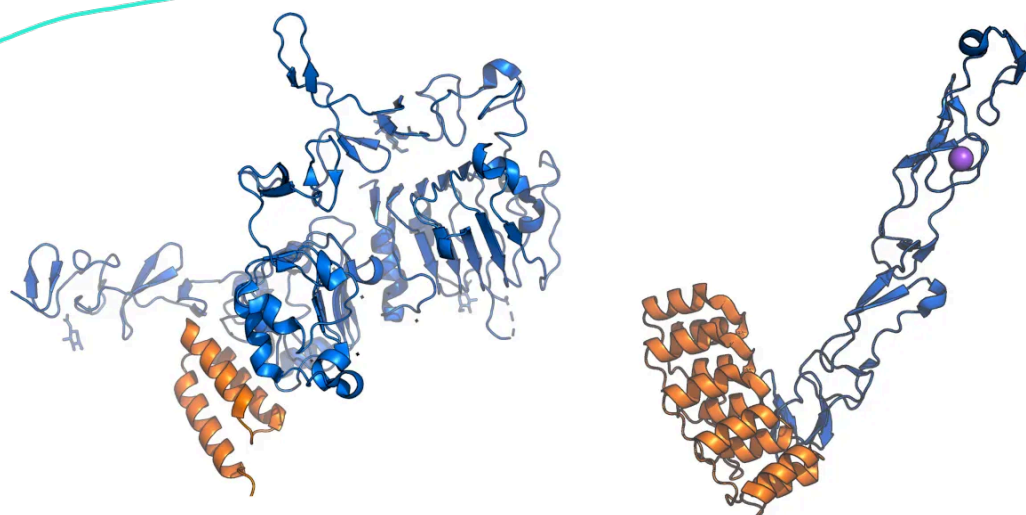
The last advantage usually cited is size. Small proteins cross endothelial junctions and transcytotic pathways more readily than large ones, and once in the tissue, they can diffuse more readily through the extracellular space due to their [small hydrodynamic radius](#). Unfortunately, that advantage is also the source of most of their problems...

Miniproteins have been around for some time

Miniproteins are not new. Despite the recent hype, they have been an active drug modality for decades, with several going through advanced stages of the clinic.

Take, for example, [affibodies](#). Several decades ago, a team at KTH characterised [Protein A](#) from *S. aureus*, a protein in the cell wall of this pathogen with the unusual ability to bind immunoglobulins. In a brilliant exercise in protein engineering, the researchers realised that their engineered “domain Z” had a binding surface that could be reprogrammed. They identified 13 positions that could be varied with

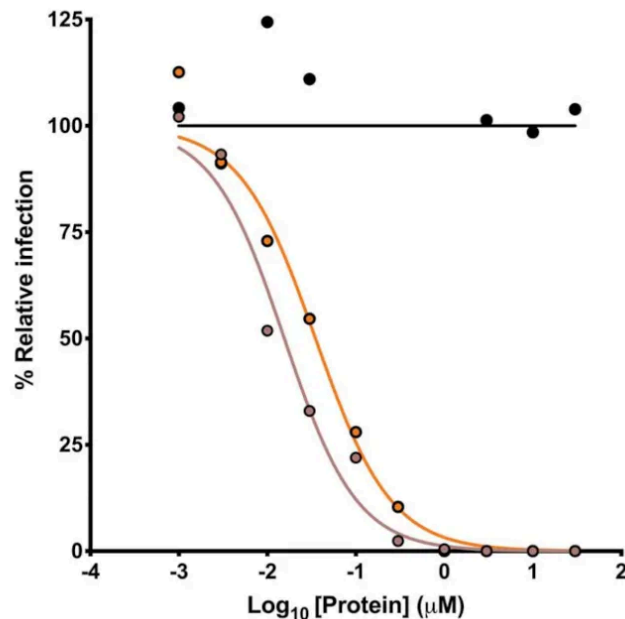
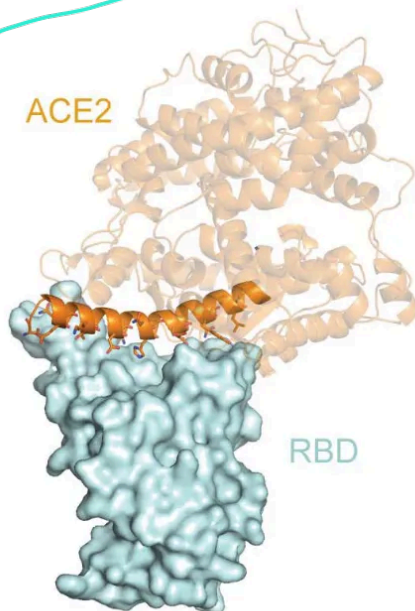
combinatorial libraries and a screening method like [phage display](#), giving a theoretical diversity of 20^{13} variants. The method can produce miniprotein binders against different targets, and has been commercialised by the Swedish company [Affibody AB](#). They have spent nearly thirty years building molecules, and their [izokibep](#) asset, a binder of [IL-17A](#), recently completed a successful Phase III clinical trial for hidradenitis suppurativa.



Crystal structures of an affibody (left) and a DARPin (right), binding some human targets. Affibodies are three-helix bundle proteins of about 60 amino acids; here (PDB: 3MZW) shown binding [HER2](#). DARPins are modular proteins built from tandem ankyrin repeats, typically about 120-180 amino acids, whose concave surface forms the binding interface; here (PDB: 7P3I) shown binding [CD40](#).

Affibodies are not the only example of the so-called “antibody mimetics”. [DARPins](#) (Designed Ankyrin Repeat Proteins) work on the same principle with a different scaffold. Each repeat is composed of a beta-turn, followed by two anti-parallel alpha-helices, stacked into a rigid, tightly packed unit. A three-repeat DARPin presents roughly 21 independently variable amino-acid positions, which can be combinatorially modified. The technology was taken to the market by [Molecular Partners](#), which has also been producing multiple assets. There are also other less well-known miniproteins like [anticalins](#) or [monobodies](#), which use different scaffolds to arrive at the same idea.

Then, the computers arrived. In the late 2010s (mind you, *before* the deep learning revolution!), the progress in protein design led to some computationally designed proteins. For example, David Baker's group generated anti-spike protein inhibitors that were able to stop CoVID-19 infections in an animal model. The team tried to take it to the clinic, but the molecule was overtaken by events: new CoVID-19 variants kept changing the spike, and it was not worth developing a protein you would have to redesign every few months.



On the left, the diagram of the designed miniproteins binding the receptor binding domain (RBD) SARS-CoV-2 spike protein, and thus blocking binding to the ACE2 receptor that the virus engages for internalisation. On the right, a neutralisation assay utilising two of the designed proteins, with a negative control (in black) on influenza. Images reproduced from Figures 1 and 5 in the [Science paper](#).

While the CoVID-19 miniproteins did not end up in the clinic, there is one designed protein that did. The lab of David Baker also produced [NL-201](#), a dual agonist of [IL-2](#) and [IL-15](#), which was taken to the clinic by the UW spin-off Neoleukin Therapeutics, in what is the first clinical trial on a fully *de novo* designed miniprotein. While the Phase I clinical trial was successful, Neoleukin discontinued the program in November 2022; we will come back to this shortly.

There is no doubt that you can design miniproteins, and that they have applications. What is less often said is that we have been putting them into people for fifteen years, and therefore have a wealth of data on how they perform in the clinic.

Are miniproteins good drugs?

Binding a protein strongly is the first step, but it is certainly not enough. You also need something you can administer to a patient, and that stays in the patient long enough to do something useful. Simplifying it a lot, drugs leave the circulation through a combination of metabolism or catabolism and excretion. Small proteins are particularly vulnerable to elimination by the kidney: as hydrodynamic size falls, [glomerular filtration](#) generally rises. The most common rule of thumb is that molecules under 60–70 kDa are quickly cleared, so a 60 to 100 amino acid miniprotein will have a blood half-life of minutes to hours.

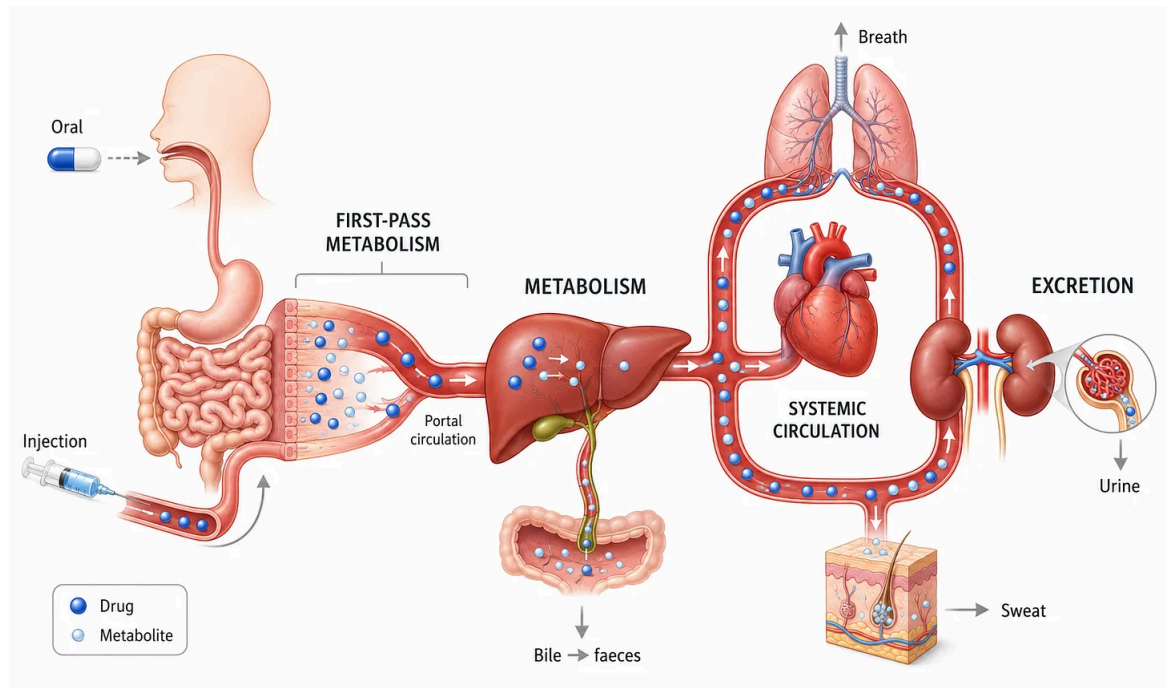


Diagram of the main elements that affect the delivery of a drug inside the organism. Proteins are generally administered by injection (although there are some proofs of concept with proteins that can survive digestion in the gut), and **have to deal with metabolism by the liver and excretion by the kidney**. While proteins are generally resistant to liver metabolism, they suffer glomerular filtration by the kidney. Image generated with GPT Image 2.

There is no surprise that **nearly all designed proteins in the clinic have been engineered with an albumin-binding domain**. The solution doesn't come for free, though. The albumin binding domain significantly increases the size of the protein, so any tissue permeation advantage immediately disappears. Albumin also changes the tissue distribution, rather than merely extending half-life, and undergoes **FcRn recycling** leading to distinct trafficking behaviour. That can be beneficial, but it means the pharmacokinetics of the therapeutic are not simply "the same miniprotein, but longer-lived". Paradoxically, **the standard fix for the central liability of a miniprotein is to make it bigger and give it some of the pharmacology of an antibody!**

This situation is very different from antibodies, where the fragment crystallizable (Fc) tends to bind to Fc receptors in blood that maintain it stable⁴. Note that **a big part of antibody engineering involves exploiting well-known mutations on the Fc that increase binding to FcRn and therefore half-life in blood.**

The other classical objection to designed miniproteins is **immunogenicity**. Brushing aside a lot of details, one of the mechanisms that our immune system uses to recognise foreign proteins is to expose peptides in the surface of cells. Peptides obtained from the constant degradation of proteins are exposed on the **major histocompatibility complex**. If those peptides activate helper T cells, they can drive anti-drug-antibody responses. *De novo* or non-human sequences may increase this risk, although the effect is not universal: several designed proteins have gone through Phase I clinical trial

showing little immunogenicity (and some antibodies have had clear problems). In particular, the only known example of a clinical trial on a *de novo* designed miniprotein, the NL-201 protein, did not show immunogenicity. But it is certainly a worrying factor in many miniprotein drugs.

So let us look at a few examples.

There is only one miniprotein that has cleared the regulatory pathway: [ecallantide](#), approved by the FDA in December 2009. It is not really a miniprotein as we commonly understand it: it is an engineered [Kunitz domain](#) (a standard protease inhibitor domain) evolved by phage display to inhibit the plasma protease [kallikrein](#), and used in the treatment of angioedema. The binder is extremely effective biochemically (K_i around 25 pM), and the clinical trials [were successful](#), but the real problem was [tolerability](#). The protein comes with a [box warning for anaphylaxis](#), restricting administration to a healthcare professional equipped to manage it. Nearly 20% of the patients assessed developed anti-ecallantide antibodies; and [many others developed adverse immune reactions](#) that may be related to the foreign protein effect we discussed earlier. The miniprotein has eventually been displaced by an antibody, [lanadelumab](#), which — tellingly — came out of [the company that had acquired its developer](#).

Another instructive case is [izokibep](#), an affibody for [IL-17A](#) that recently completed a successful Phase III trial (after a very tricky Phase II trial [allegedly botched by an untrustworthy CRO](#)... the stuff of nightmares). This should be exciting: succeeding all the way through Phase III is something to be truly proud of. However, despite the positive results, izokibep is not currently starting commercialisation. ACELYRIN had [licensed the molecule in 2021](#) from Affibody AB, but after the trial completion they [announced a retreat](#), claiming that “it was best brought to market by a larger organisation”. Assets with a positive Phase III in a blockbuster class do not usually sit on the shelf for nearly two years — this is likely a signal that antibody competitors of izokibep ([secukinumab](#), [bimekizumab](#), etc) are significantly better. The miniprotein did its job very well, but it turns out that it had no advantage with respect to antibodies.

The final example I will look into is the first fully *de novo* designed protein to enter the clinic: Neoleukin's NL-201. Descended from [the Baker lab's Neo-2/15](#), it is a mimic of IL-2 and IL-15 that binds the [IL-2R \$\beta\$ receptor](#) more tightly than the natural ligands ($K_d \approx 19$ nM for the published parent). The Phase I clinical trial was very successful, showing target engagement in solid tumours, and good pharmacodynamics. More interestingly, it did not show significant immunogenicity even after multiple cycles of dosing, which is the most commonly expected risk for *de novo* protein therapeutics. Nevertheless, Neoleukin discontinued it on benefit-risk in a crowded IL-2 landscape,

cut 40% of staff, and wound the company down. It failed for none of the reasons a protein designer would have predicted.

These are the most advanced miniproteins ever developed, and unfortunately, every single one of them failed as drugs. Ecallantide reached the market with serious immunogenicity liabilities, and was displaced by an antibody from the company that bought its developer. Izokibep has been abandoned despite clearing Phase III. NL-201 achieved nearly everything a designed protein was supposed to do and was shut down anyway. Note that none of these proteins failed because they were not good binders, but because of tolerability and differentiation against antibodies.

If you got this far, you enjoyed my essay at least a little. Subscribe (free) for more essays on frontier AI in the life sciences.

4.

Where the field is converging: radiopharmaceuticals

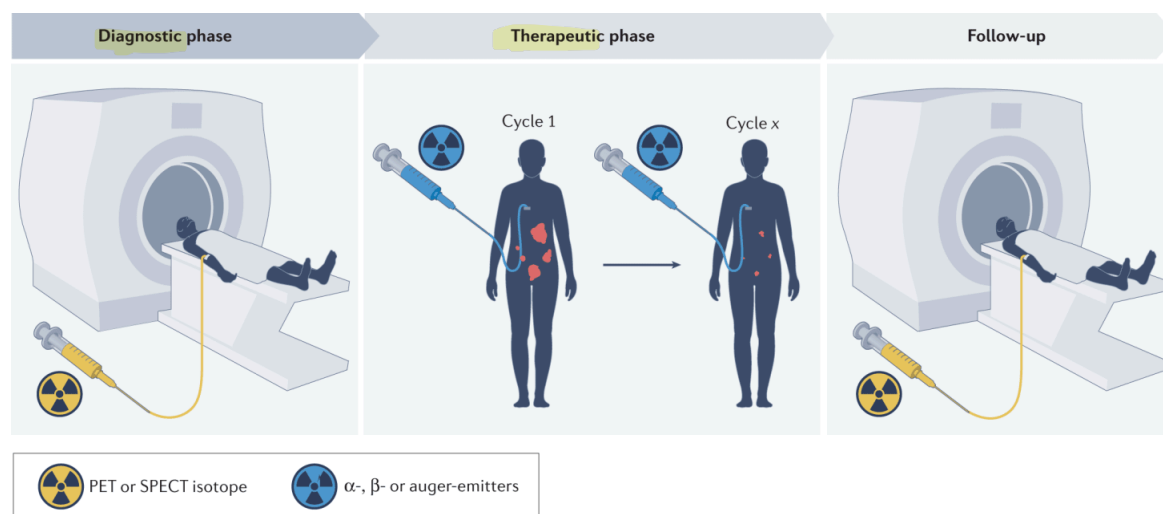
If you want to know what protein design companies actually believe about miniproteins as drugs, go and check what they are doing with their pipelines. Nearly all traditional miniprotein companies are now pivoting in the same direction: **radiotheranostics**.

Affibody AB? They abandoned one program after Phase I, and they are still waiting for someone to partner with to commercialise their [izokibep](#) molecule currently in Phase III. Their webpage now features “*the next generation radiopharmaceuticals*”, very different from what they had [just a year earlier](#). Molecular Partners? They have abandoned their purist DARPin projects, and instead they are doubling down on radiopharmaceuticals. Even [Bicycle Therapeutics](#), whose value proposition has focused on “*combining the pharmacology usually associated with a biologic with the manufacturing and pharmacokinetic properties of a small molecule*”, is now [pivoting to radiotherapeutics](#).

So, why this, and why all at once?

Strictly speaking, a **theranostic** is a type of therapeutic that can be used both *treat and diagnose* a disease. By and large, theranostics have been radiopharmaceuticals, because the chemistry cooperates: there are well-characterised isotope pairs with very similar geometric properties but completely different nuclear behaviour. For example, ^{68}Ga , a positron emitter that can be used for [PET scans](#), and ^{177}Lu , an emitter of [beta particles](#) that has localised cytotoxic effects, bind to the same chelator moiety ([DOTA](#)). This leaves a very interesting therapeutic approach where you can first use one nuclide

to validate that a target is prominent in a cancer patient, and then hit it with a cytotoxic variant.



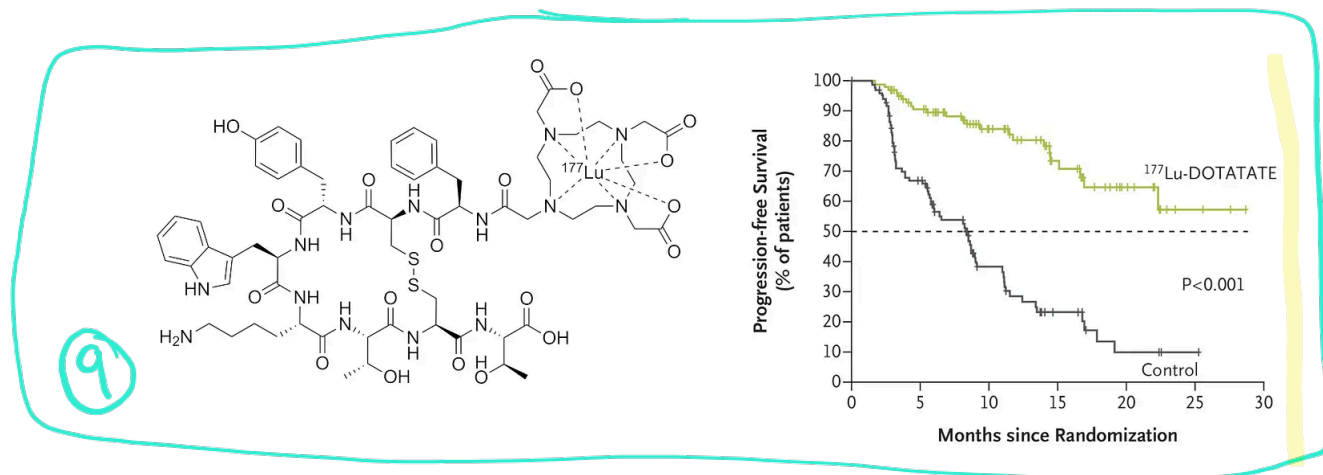
A diagram of the steps in radiotheranostic treatment ([source](#)). The patient is first treated with a radionuclide that can be detected in a PET or SPECT scan, verifying that it has the expression pattern conducive to successful treatment. The treatment phase swaps the radionuclide for one that can destroy cells. Tumour reduction can be followed up using the same diagnostic method.

I like to think of theranostics as a variant of antibody-drug conjugates, a therapeutic modality where a cytotoxic molecule is bound to an antibody that directs it to a tissue. The big difference is that an ADC's payload is at least nominally conditional: linker chemistry, protease sensitivity, pH lability and internalisation all give the designer levers over when and where the toxin is released. A radionuclide has none of that. It decays on a schedule set by nuclear physics, indifferent to where it is, and no chemistry can stop it. The only variable you control is where the molecule is standing when the atom goes off. That makes the tumour-to-normal-tissue residence ratio the entire ballgame, and it is why a vector that leaves the bloodstream in hours but sits in a tumour for days is the ideal shape. This is exactly what is best for miniproteins.

For a radiotheranostic, a small protein that penetrates tissue quickly and is cleared by the kidney within hours is not a compromise — it is the desired format!

The radiotheranostics revolution started with [Lutathera](#)⁵ (developed by Novartis, approved in 2018 and currently generating some \$1B in annual revenue). The somatostatin receptor SSTR2 is a common lineage marker in neuroendocrine cancers. Therapeutic approaches using somatostatin variants such as octreotide, engineered for protease resistance, have been used as imaging agents for decades. The Lutathera idea was to change the traditional radionuclide⁶ for a DOTA moiety that could exchange between ⁶⁸Ga and ¹⁷⁷Lu. What makes it work is that SSTR2 internalises on binding, and once inside the lysosome the peptide is degraded but the radiometal chelate cannot escape — so unbound drug clears renally in hours while the tumour holds its

dose for days, and the ^{177}Lu betas deposit their energy over roughly 0.7 mm, a few cell diameters, killing labelled cells and their unlabelled neighbours alike.



Take [Aktis Oncology](#) (NASDAQ: AKTS), for example. Their AKY-1189 program is very compelling: they take a protein, [nectin-4](#), that is highly overexpressed in metastatic urothelial carcinoma (mUC) as well as other solid tumours. They developed a small miniprotein binder and loaded it with actinium-225, a potent emitter of alpha particles. The molecule entered a Phase Ib trial in May 2025, and [received FDA Fast Track designation](#). But the most interesting part of the story is the indication: nectin-4 already has an approved antibody drug ([enfortumab vedotin](#))! The miniprotein is not merely reproducing antibody pharmacology: its faster distribution and blood clearance provide a more favourable dosimetric profile for a radioactive payload.

(ADC)

The pivot is not free. Radiopharmaceuticals come with their own constraints: for example, isotope half-lives run from days down to about an hour for ^{68}Ga , so proximity to a generator or a reactor is not a logistics detail but a strategic one. Indeed, Novartis, the maker of Lutathera, has [invested heavily in radiopharmaceutical manufacturing facilities](#) across the US. The development of novel protein binders will be useful but it is most likely not the main problem.

And that is the tell. Several companies have spent decades building protein binder platforms. And they work, producing excellent binders, reliably, against hard targets — but they have not produced a competitive drug. So the companies have moved to the one application where miniproteins beat antibodies outright, and that application is defined by their central liability becoming an asset. And this was not a choice: they arrived at it by elimination.

The prospects of designed proteins

In my opinion, miniproteins are the latest victim of the life sciences overhyping disease. The pattern is always the same: an exciting technology arrives, and the first thing anyone asks is *what drug it can make*. Yes, therapeutics is the most impactful

thing you can do in this industry (and, cynically, where most of the money is), but it is also the hardest possible place to start.

This brings me back to where I began. The Anthropic release presents designing new miniproteins as a “proxy” for drug discovery. I disagree. A proxy is only useful if performance on it predicts performance on the target task, and binder-design success predicts essentially nothing about PK, tox, target validation, or clinical translation, which is where drug discovery actually fails. Every molecule in the previous section cleared the binder bar with flying colours and died somewhere else.

I want to make something clear, though: the scientific achievements of designed proteins are nothing short of remarkable. I just don't think that drugs are the best application.

Start with diagnostics and biosensors, where stability in plasma is not an issue. Here miniproteins are a straight substitution for antibodies: cheaper, more stable, and equally effective as a detection mechanism. And this is not limited to detecting proteins, since it is also relatively easy to [design artificial miniproteins that bind small molecules](#) with great specificity. There are clear applications in vaccine development too, as a way to expose a pathogen epitope in the right conformation with a synthetic protein — and there the immunogenicity that sinks a therapeutic is the entire mechanism. And I still think that de novo enzymes catalysing novel chemical reactions are one of the most exciting fronts in the whole of computational biology. Notice what all of these have in common: not one of them requires a molecule to survive in blood.

These alternative verticals are not only better suited to the strengths of miniproteins, but they are also faster to get to revenue and require much smaller investment. Whereas therapeutics require 10-15 years of clinical development, a diagnostic reagent can be sold within a few years of designing it, and an enzyme can be licensed to a manufacturer almost immediately after proving efficacy. If designed proteins are going to prove their worth, this is where it will happen first — and possibly where it matters most.

Conclusions

Designed protein binders are a truly outstanding achievement, but likely not the harbinger of new drugs. Every major miniprotein company is now cutting down their traditional projects and betting everything on radiopharmaceuticals; they wouldn't be doing this if having a binder were such a valuable thing.

The problem is that miniprotein binders will always compete with antibodies. In this space, miniproteins find themselves with worse footing, not just because they need to engineer the elements that antibodies take for granted (relatively low immunogenicity,

stability in plasma, etc), but also because antibodies have had decades to go after all the usual targets. Because of the crowding of the space, and the struggle to differentiate, it is no surprise that the only remaining space for miniproteins is the one where antibodies cannot compete: the delivery of a radioactive payload where rapid clearance is a must.

None of this means miniproteins are a dead end. The applications I described above — diagnostics, biosensors, vaccine scaffolds, *de novo* enzymes — do not suffer from the liabilities that have blocked miniprotein progress as drugs. There also may well be situations where a designed protein has an enormous impact as a drug. [Omomyc](#), which we have not discussed in this essay, breaks nearly all of the assumptions above: it is internalised, and it targets the extremely undruggable and even more sought-after target [Myc](#). Omomyc was the fruit of serendipity, but we may one day be able to design proteins that achieve the same by design rather than by luck.

But until we figure out the elements of *drug* discovery, then miniprotein binders will not be good drugs. As a benchmark for protein design, they fill my heart with joy. As a proxy for drug discovery, they are the TikZ unicorn of biology.

Thanks for reading! Subscribe for free if you would like to read more essays on frontier AI and the life sciences.

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- 1 Provided, of course, that we implement the appropriate guardrails for biosecurity. This is too large a topic to squeeze in this essay, and will be writing more on it in the future.
 - 2 This size is enough to fall under FDA's definition ([21 CFR § 600.3\(h\)\(6\)](#)) that a protein is “any alpha amino acid polymer with a specific, defined sequence that is greater than 40 amino acids in size”, and therefore miniproteins are *biologics*.
 - 3 The cost of goods sold (COGS) is [only one of the contributors](#) to the exorbitant price of antibodies, which is something I want to talk about in the future.
 - 4 An additional advantage of a conventional IgG is bivalency: its two Fab arms each contain an antigen-binding site, allowing [avidity](#) when both engage the same target or surface.
 - 5 Radiotheranostics have been in use for much longer, but Lutathera is the protein-based example that started the current shift.
 - 6 The original octreotide was labelled with ¹¹¹In and used a different chelator.
-

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...

Thanks Carlos, another great one! You wrote about the pivot from mini proteins to radioligand therapies but I don't think you actually gave your opinion. Do you think radioligand therapies will be revolutionary? Are they going to improve outcomes for a bunch of cancers? Why or why not?

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